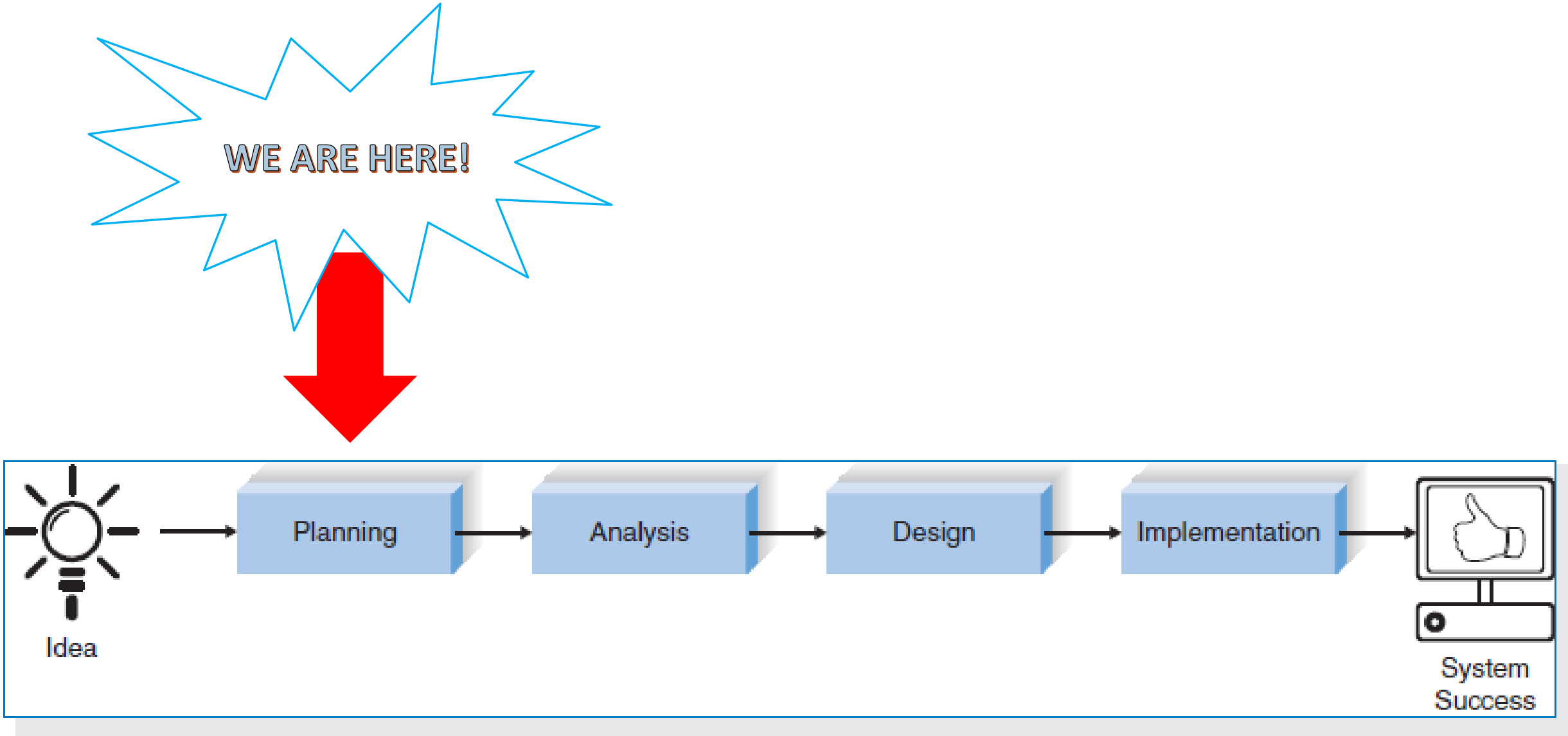
System Analysis

DR. DRAGUNOV

3_1 Project Initiation


WE ARE HERE!
Idea
Planning
Analysis
Design
Implementation
System
Success

SDLC: The Planning Phase

1. Project Initiation

- Develop/receive a system request
- Conduct a feasibility analysis
  - Technical feasibility
  - Economic feasibility
  - Organizational feasibility

2. Project Management

- Develop the work plan
- Staff the project
- Monitor & control the project

PROJECT SELECTION AND MANAGEMENT



PROJECT IDENTIFICATION AND INITIATION

- Where do project ideas come from?

A project is identified when someone in the organization identifies a business need to build a system.

- Could you give me examples of such needs?

Examples of business needs include supporting a new marketing campaign, reaching out to a new type of customer, or improving interactions with suppliers. Sometimes, needs arise from some kind of “pain” within the organization, such as a drop in market share, poor customer service levels, unacceptable product defect rates, or increased competition. New business initiatives and strategies may be created and a system to support them is required, or a merger or acquisition may require systems to be integrated.

PROJECT IDENTIFICATION AND INITIATION

Business process management (BPM)

- methodology used by organizations to continuously improve end-to-end business processes. Business process management can be applied to internal organizational processes and to processes spanning multiple business partners.

Benefits:

■ enhanced process agility, giving the organization the ability to adapt more rapidly and effectively to a changing business environment;

■ improved process alignment with industry “best practices”; and

■ increased process efficiencies as costs are identified and eliminated from process workflows.

PROJECT IDENTIFICATION AND INITIATION

BPM generally follows a continuous cycle of systematically creating, assessing, and altering business processes. Business analysts, with their in-depth business knowledge, play a particularly important role in business process management by:

1. Defining and mapping the steps in a business process,

2. Creating ways to improve on steps in the process that add value,

3. Finding ways to eliminate or consolidate steps in the process that don't add value,

4. Creating or adjusting electronic workflows to match the improved process maps.

The last step is particularly relevant to our discussion since the need for information systems projects is frequently identified here.

PROJECT IDENTIFICATION AND INITIATION

Business Process Automation

In fact, the automation of business processes (termed *Business Process Automation*), is the foundation of many information technology systems. In these situations, technology components are used to complement or substitute for manual information management processes with the intent of gaining cost efficiencies.

PROJECT IDENTIFICATION AND INITIATION

Business Process Improvement results from studying the business processes, creating new, redesigned processes to improve the process workflows, and/or utilizing new technologies enabling new process structures (steps 2, 3, and 4 above).

PROJECT IDENTIFICATION AND INITIATION

Business Process Management may also reveal the need for the complete revamping of the organization's business processes, termed

Business Process Reengineering (BPR)

BPR means changing the fundamental way in which the organization operates—"obliterating" the current way of doing business and making major changes to take advantage of new ideas and new technology. As you might expect, BPR projects involve substantial risk due to the significant organizational and operational changes that result. Top management support and careful management are critical in these fairly rare types of projects.

Both IT people (i.e., the information systems experts) and business people (i.e., the subject matter experts) should work closely together to find ways for technology to support business needs.

PROJECT IDENTIFICATION AND INITIATION

The business need drives the high-level business requirements for the system. Business requirements describe the reasons for developing the system and outline the benefits it will provide the organization.

Once the project sponsor identifies a project that meets an important business need and he or she can identify the business requirements and business value of the system, it is time to formally initiate the project. In most organizations, project initiation begins by preparing a *system request*.

System Request

A *system request* is a document that describes the business reasons for building a system and the value that the system is expected to provide. The project sponsor usually completes this form as part of a formal system project selection process within the organization.

Most system requests include five elements:

- project sponsor,
- business need,
- business requirements,
- business value, and
- special issues.

System Request

Element	Description	Examples
Project Sponsor	The person who initiates the project and who serves as the primary point of contact for the project on the business side	Several members of the finance department Vice president of marketing IT manager Steering committee CIO CEO
Business Need	The business-related reason for initiating the system	Increase sales Improve market share Improve access to information Improve customer service Decrease product defects Streamline supply acquisition processes

System Request

Business Requirements	The business capabilities that the system will provide	Provide online access to information Capture customer demographic information Include product search capabilities Produce management reports Include online user support
Business Value	The benefits that the system will create for the organization	3% increase in sales 1% increase in market share Reduction in headcount by 5*FTEs $200,000 cost savings from decreased supply costs $150,000 savings from removal of existing system

System Request

Special Issues or Constraints	Issues that are relevant to the implementation of the system that need to be known by the approval committee	Government-mandated deadline for May 30 System needed in time for the Christmas holiday season Top-level security clearance needed by project team to work with data

Feasibility Analysis

During *project initiation*, the system's business value to the organization is identified — how will it contribute to the organization's future success?

A system request presents a brief summary of a business need and explains how a system that addresses the need will create business value.

The feasibility analysis examines key aspects of the proposed project:

- The technical feasibility (Can we build it?)
- The economic feasibility (Will it provide business value?)
- The organizational feasibility (If we build it, will it be used?)

The *system request* and *feasibility analysis* are presented to an information systems approval committee (sometimes called a steering committee), which decides whether the project should be undertaken.

Feasibility Analysis

Once the need for the system and its business requirements have been defined, the approval committee may authorize the systems analyst to prepare a more detailed business case to better understand the proposed information system project. ***Feasibility analysis*** guides the organization in determining whether to proceed with the project.

Feasibility analysis also identifies the important *risks* associated with the project that must be managed if the project is approved.

Feasibility Analysis

Technical Feasibility: Can We Build It?

- Familiarity with application: Less familiarity generates more risk.
- Familiarity with technology: Less familiarity generates more risk.
- Project size: Large projects have more risk.
- Compatibility: The harder it is to integrate the system with the company's existing technology, the higher the risk will be.

Feasibility Analysis

Economic Feasibility: Should We Build It?

- Development costs
- Annual operating costs
- Annual benefits (cost savings and/or increased revenues)
- Intangible benefits and costs

Feasibility Analysis

Organizational Feasibility: If We Build It, Will They Come?

- Project champion(s)
- Senior management
- Users
- Other stakeholders
- Is the project strategically aligned with the business?

Technical Feasibility

The first technique in the feasibility analysis is to assess the *technical feasibility* of the project, the extent to which the system can be successfully designed, developed, and installed by the IT group. Technical feasibility analysis is, in essence, a *technical risk analysis* that strives to answer the question: "***Can we build it?***"

Economic Feasibility

The second element of a feasibility analysis is to perform an *economic feasibility* analysis (also called a *cost–benefit analysis*). This attempts to answer the question "***Should we build the system?***" Economic feasibility is determined by identifying costs and benefits associated with the system, assigning values to them, calculating future cash flows, and measuring the financial worthiness of the project.

Keep in mind that organizations have limited capital resources and multiple projects will be competing for funding. The more expensive the project, the more rigorous and detailed the analysis should be.

Development Costs	Operational Costs
Development team salaries	Software upgrades
Consultant fees	Software licensing fees
Development training	Hardware repairs
Hardware and software	Hardware upgrades
Vendor installation	Operational team salaries
Office space and equipment	Communications charges
Data conversion costs	User training

Tangible Benefits	Intangible Benefits
Increased sales	Increased market share
Reductions in staff	Increased brand recognition
Reductions in inventory	Higher quality products
Reductions in IT costs	Improved customer service
Better supplier prices	Better supplier relations

Cash Flow Analysis and Measures

IT projects commonly involve an initial investment that produces a stream of benefits over time, along with some ongoing support costs. Therefore, the value of the project must be measured over time. Cash flows, both inflows and outflows, are estimated over some future period. Then, these cash flows are evaluated using several techniques to judge whether the projected benefits justify incurring the costs.

A very basic cash flow projection is shown in Figure (next slide)

Cash Flow Analysis and Measures

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	57,000	152,000
Total Costs	100,000	10,000	12,000	16,000	138,000
Net Benefits (Total Benefits − Total Costs)	(100,000)	35,000	38,000	41,000	14,000
Cumulative Net Cash Flow	(100,000)	(65,000)	(27,000)	14,000	

Two of the common methods for evaluating a project's worth can now be determined. Each of these calculations will be explained here:

Return on Investment

The return on investment (ROI) is a calculation that measures the average rate of return earned on the money invested in the project. ROI is a simple calculation that divides the project's net benefits (total benefits total costs) by the total costs. The ROI formula is:

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$

$$\text{ROI} = \frac{152{,}000 - 138{,}000}{138{,}000} = \frac{14{,}000}{138{,}000} = 10.14\%$$

A high ROI suggests that the project's benefits far outweigh the project's cost, although exactly what constitutes a "high" ROI is unclear. ROI is commonly used in practice; however, it is hard to interpret and should not be used as the only measureof a project's worth.

Break-Even Point

The *break-even point* (also called the *payback method*) is defined as the number of years it takes a firm to recover its original investment in the project from net cash flows.

(In the year in which Cumulative Cash Flow turns positive):

$$\text{BEP} = \begin{array}{l}\text{Number of}\\ \text{years of}\\ \text{negative}\\ \text{cash flow}\end{array} + \frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$$

$$\text{BEP} = 2 + \frac{41{,}000 - 14{,}000}{41{,}000} = 2 + \frac{28{,}000}{41{,}000} = 2.68 \text{ years}$$

Discounted Cash Flow Technique

The simple cash flow projection shown in Figure, and the return on investment and break-even point calculations all share the weakness of not recognizing the time value of money. In these analyses, the timing of cash flows is ignored. A dollar in Year 3 of the project is considered to be exactly equivalent to a dollar received in Year 1.

Discounted cash flows are used to compare the present value of all cash inflows and outflows for the project in today's dollar terms. The key to understanding present values is to recognize that if you had a dollar today, you could invest it and receive some rate of return on your investment. Therefore, a dollar received in the future is worth less than a dollar received today, since you forgo that potential return. Assuming you could have invested $100 at a 10% rate of return, you'll be receiving the equivalent of $75 in today's terms.

Discounted Cash Flow Technique

The basic formula to convert a future cash flow to its present value (PV) is:

$$PV = \frac{\text{Cash flow amount}}{(1 + \text{rate of return})^n}$$ where n is the year in which the cash flow occurs.

$$PV = \frac{100}{(1 + .10)^3} = \frac{100}{1.331} = 75.13$$

Discounted Cash Flow Technique

In this figure, the present value of the projected benefits and costs shown in previous figure have been calculated using a 10% required rate of return.

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	55,000	
PV of Total Benefits		40,909	41,322	42,825	125,056
Total Costs	100,000	10,000	12,000	16,000	
PV of Total Costs	100,000	9,091	9,917	12,021	131,029

Net Present Value (NPV)

The NPV is simply the difference between the total present value of the benefits and the total present value of the costs.

$$\begin{aligned} \text{NPV} &= \sum \text{PV of Total Benefits} - \sum \text{PV of Total Costs} \\ &= \$125{,}056 - \$131{,}029 = (\$5{,}973) \end{aligned}$$

Return on Investment

As long as the NPV is greater than zero, the project is considered economically acceptable. Unfortunately for this project, the NPV is less than zero, indicating that for a required rate of return of 10%, this project should not be accepted. The required rate of return would have to be something less than ???% before this project returns a positive NPV. This example illustrates the fact that sometimes the “naïve” techniques of ROI and BEP find that the project appears acceptable, but the more rigorous and financially correct NPV technique finds the project is actually unacceptable.

Answer 6.65;

Figure reviews the steps involved in performing an economic feasibility analysis.

1. Identify Costs and Benefits	List the tangible costs and benefits for the project. Include both one-time and recurring costs.
2. Assign Values to Costs and Benefits	Work with business users and IT professionals to create numbers for each of the costs and benefits. Even intangibles should be valued if at all possible.
3. Determine Cash Flow	Forecast what the costs and benefits will be over a certain period, usually, three to five years. Apply a growth rate to the values, if necessary.

4. Assess Project's Economic Value	Evaluate the project's expected returns in comparison to its costs. Use one or more of the following evaluation techniques:
• **Return on Investment (ROI)**	Calculate the rate of return earned on the money invested in the project, using the ROI formula.
• **Break-Even Point (BEP)**	Find the year in which the cumulative project benefits exceed cumulative project costs. Apply the breakeven formula, using figures for that year. This calculation measures how long it will take for the system to produce benefits that cover its costs.
• **Net Present Value (NPV)**	Restate all costs and benefits in today's dollar terms (present value), using an appropriate discount rate. Determine whether the total present value of benefits is greater than or less than the total present value of costs.

Organizational Feasibility

The final technique used for feasibility analysis is to assess the organizational feasibility of the system: how well the system ultimately will be accepted by its users and incorporated into the ongoing operations of the organization. There are many organizational factors that can have an impact on the project, and seasoned developers know that organizational feasibility can be the most difficult feasibility dimension to assess. In essence, an organizational feasibility analysis attempts to answer the question “If we build it, will they come?”

Organizational Feasibility

One way to assess the organizational feasibility of the project is to **understand how well the goals of the project align with business objectives**. Strategic alignment is the fit between the project and business strategy—the greater the alignment, the less risky the project will be, from an organizational feasibility perspective.

A second way to assess organizational feasibility is **to conduct a stakeholder analysis.** A stakeholder is a person, group, or organization that can affect (or can be affected by) a new system. In general, the most important stakeholders in the introduction of a new system are the **project champion**, system users, and organizational management.

The *champion* is a high-level executive and is usually, but not always, the project sponsor who created the system request. The champion supports the project by providing time and resources (e.g., money) and by giving political support within the organization by communicating the importance of the system to other organizational decision makers. More than one champion is preferable because if the champion leaves the organization, the support could leave as well.

Role		To Enhance Organizational Feasibility
Champion	A champion: • Initiates the project • Promotes the project • Allocates his or her time to the project • Provides resources	• Make a presentation about the objectives of the project and the proposed benefits to those executives who will benefit directly from the system. • Create a prototype of the system to demonstrate its potential value.
Organizational Management	Organizational managers: • Know about the project • Budget enough money for the project • Encourage users to accept and use the system	• Make a presentation to management about the objectives of the project and the proposed benefits. • Market the benefits of the system, using memos and organizational newsletters. • Encourage the champion to talk about the project with his or her peers.
System Users	Users: • Make decisions that influence the project • Perform hands-on activities for the project • Ultimately determine whether the project is successful by using or not using the system	• Assign users official roles on the project team. • Assign users specific tasks to perform, with clear deadlines. • Ask for feedback from users regularly (e.g., at weekly meetings).

SUMMARY

Project Identification and Initiation

System Request

Feasibility Analysis

Small practice

During Project Planning phase, next cash flows were identified:

Hardware repairs: $30 000

Increased sales: $20 000

Reduction in staff: $40 000

Development team salaries: $ 50 000

Office space and equipment: $10 000

Data conversion tools: $10 000

Find the year then cumulative net cash flow turns to positive

Small practice

During Project Planning phase, next cash flows were identified:

Hardware repairs: $20 000

Increased sales: $40 000

Reduction in staff: $40 000

Development team salaries: $ 50 000

Office space and equipment: $10 000

Data conversion tools: $10 000

Find: ROI, BEP and NPV, if project life cycle is 2 years and rate of return = 10 %